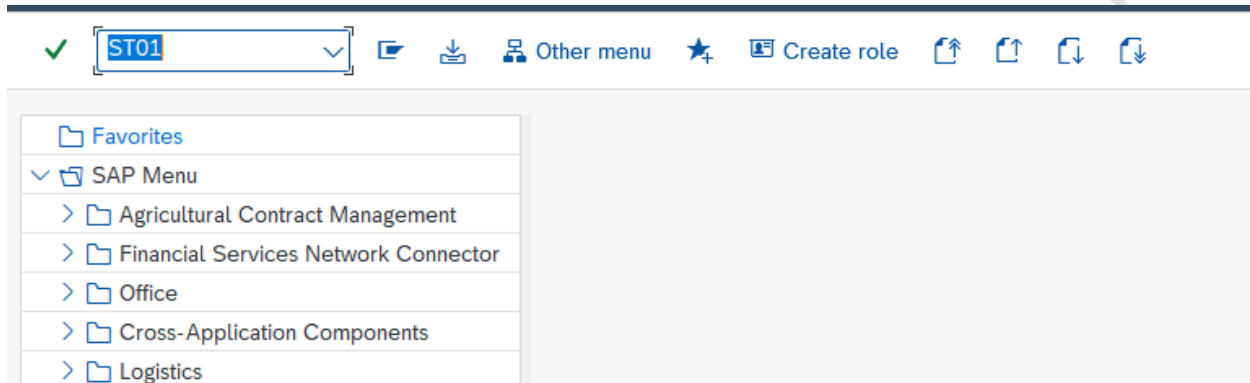


ST01 = System Trace

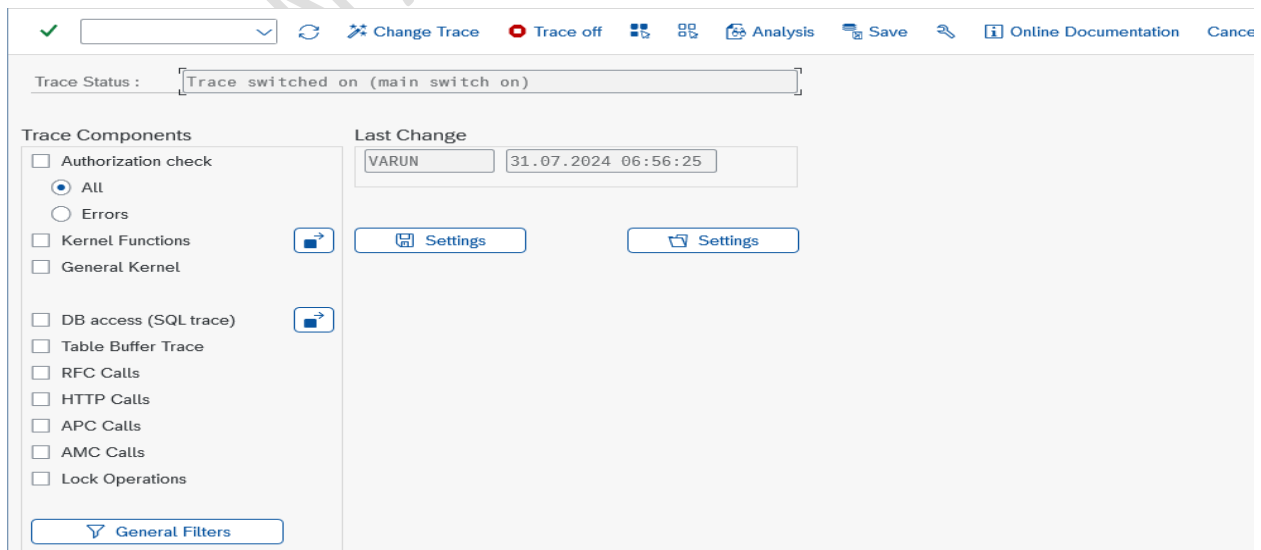
1) Access **the ST01** Transaction: -

- ❖ Go to the SAP Easy access **screen**.
- ❖ Enter the transaction code **ST01** in the command field and press **Enter**.



2) Select Trace Components: -

- In **the initial screen**, you will see options to select the type **of trace** you want **to activate**. Common options include:
 - ❖ **Authorization Check** = > To monitor authorization checks.
 - ❖ **SQL Trace** = > To log database queries.
 - ❖ **Buffer Trace** = > To monitor buffer activities.
- Check the **boxes** for the components you want to **trace**.



3) Set Trace Options (Optional): -

- You can **set filters** in the down what all are the **trace records**, such as:
 - ❖ **User IDs** = > Specify particular users to trace.
 - ❖ **Transaction Codes** = > Focus on specific transactions.
 - ❖ **Programs** = > Monitor specific programs.

System Trace: General Filters

Lock Without Reference to Trace Type

Trace for process number only	
Trace for user only	VARUN
Trace for transaction only	
Trace for program only	
Trace for RFC Module Only	
Trace for URL Only	

(either or)

✓ ✗

4) Activate the Trace: -

- ❖ Click on the "**Trace on**" button. The system will **start recording** activities based on your **selected** components and filters.

✓ [dropdown] [refresh] [Trace on] [Trace off] [Trace on] [Trace off] [Analysis] [Save] [Online Documentation] [Cancel]

5) Perform the Activities to be Traced: -

- ❖ While the trace **is active**, login to that **User id** and perform the **actions** or **processes** you want to analyze.

6) Deactivate the Trace: -

- ❖ Once you have **captured** the necessary **information**, return to the ST01 screen, and click "**Trace off**" to stop recording.

7) Display and Analyze Trace Results: -

- ❖ Click on "**Analysis**" in ST01 **screen** and specify the user, date, and time to view the **recorded data**. Then execute to **see the Trace**.

General restrictions

User name:

Client:

Work Process:

Transaction:

Duration (>us):

Max. No. of Records

From: /

To: /

Trans ID

Passport

Full Context ID:

connection ID:

Trace Records

☒ Authorization check

☒ All
☐ Errors

☒ Kernel Functions
☒ General Kernel

☒ DB access (SQL trace)
☒ Table buffer trace

☒ RFC calls
☒ HTTP Calls

☒ APC Calls
☒ AMC Calls

☒ Lock operations

8) Troubleshoot and Analyze:

Client: 812 User: VARUN Transaction: SESSION_MANAGER Trans ID: 9E50DC0D35760000E00666FFC18163BE Start: 31.07.2024 07:31:14,125951 Finish: 31.07.2024 07:31:14,133065 No. of Records: 177 File Version: 3 EPP Full Context ID: 000C29A7FD041EDF93DF268F4C0B3186 EPP Connection ID: 00000000000000000000000000000000 EPP Call Counter: 0 Block Version: 54.200 First Block of Dialog Step Last Block in Dialog Step Work Process: 9 Process ID: 10.159				
hh:mm:ss:ms	Type	Lasts(us)	Object	Text
07:31:14,132	User		@(#) \$Id: //bas/	Row: 1.796 Comment: SET_PF_STATUS: SAPLSMTR_NAVIGATION SESSION_A
Client: 812 User: VARUN Transaction: SESSION_MANAGER Trans ID: 9E50DC0D35760090E0066680631A99E2 Start: 31.07.2024 07:31:18,090265 Finish: 31.07.2024 07:31:18,122779 No. of Records: 1.882 File Version: 3 EPP Full Context ID: 000C29A7FD041EDF93DF268F4C0B3186 EPP Connection ID: 00000000000000000000000000000000 EPP Call Counter: 0 Block Version: 512.376 First Block of Dialog Step Last Block in Dialog Step Work Process: 9 Process ID: 10.159				
hh:mm:ss:ms	Type	Lasts(us)	Object	Text
07:31:18,100	AUTH		S_TCODE RC=0	tcode=SM36;TCD=SM36;type=TR;name= ;reason3=X;contextid=00007000000C29A7FD041EDF93DF
07:31:18,103	User		@(#) \$Id: //bas/	Row: 1.796 Comment: SET_PF_STATUS: MENUSYST MEN
07:31:18,105	AUTH		S_BTCH_ADM RC=0	tcode=SM36;BTCADMIN=Y;type=TR;name=SM36;reason3=X;contextid=00007000000C29A7FD041ED
07:31:18,106	AUTH		S_BTCH_JOB RC=0	tcode=SM36;JOBGROUP=' ';JOBACTION=RELE;type=TR;name=SM36;reason3=X;contextid=000070
07:31:18,108	User		@(#) \$Id: //bas/	Row: 1.796 Comment: SET_PF_STATUS: MENUSYST MEN
07:31:18,109	AUTH		S_BTCH_ADM RC=0	tcode=SM36;BTCADMIN=A;type=TR;name=SM36;reason3=X;contextid=00007000000C29A7FD041ED
07:31:18,114	AUTH		P_RUNTIME RC=0	tcode=SM36;P_APP_ID=SM36;P_ACTVT_FW=01;P_ACTVT_RT=01;type=TR;name=SM36;reason3=X;co
07:31:18,114	AUTH		P_RUNTIME RC=0	tcode=SM36;P_APP_ID=SM36;P_ACTVT_FW=01;P_ACTVT_RT=07;type=TR;name=SM36;reason3=X;co
07:31:18,114	AUTH		P_RUNTIME RC=0	tcode=SM36;P_APP_ID=SM36;P_ACTVT_FW=01;P_ACTVT_RT=04;type=TR;name=SM36;reason3=X;co

End of trace files reached.

- ❖ Use the **trace data** to identify **issues**, such as unauthorized access attempts, slow-performing SQL queries, or other system problems.

- ❖ Based on that the **security consultants** give the authorization, if fails by identify by using the **return codes in red color**.

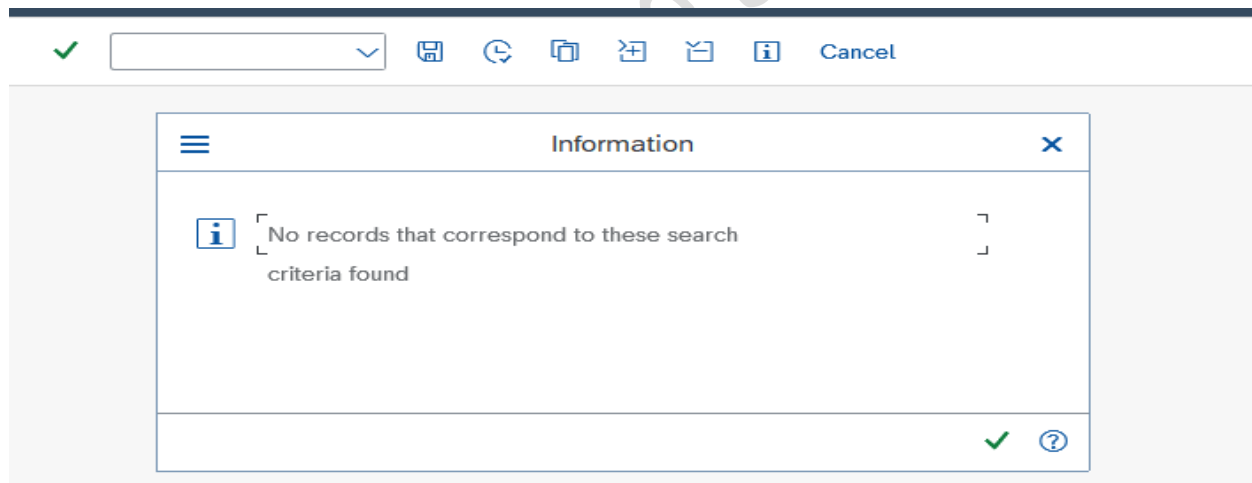
Note = > It is important to **deactivate** the trace when not in use.

Return Codes

- **RC = > 0** --- **Authorization** is successful.
- **RC = > 4** --- User has required **authorization object**, but different **authorization Values**.
- **RC = > 8** --- User does not have required **authorization values** in the user buffer.
- **RC = > 12** --- User does not have **access to authorization object**.

If any 0 Records found

→ means if no trace display.



- ✓ **Incorrect** input values.
- ✓ User present in different **application server**.

Note = > **ST01 trace** should be applied in the **local server only**. Hence before applying **ST01 trace**, ensure that **user** and **you** are in the same system.

AL08 = > Users in each server.

SM51 = > List of application servers.

STAUTHTRACE

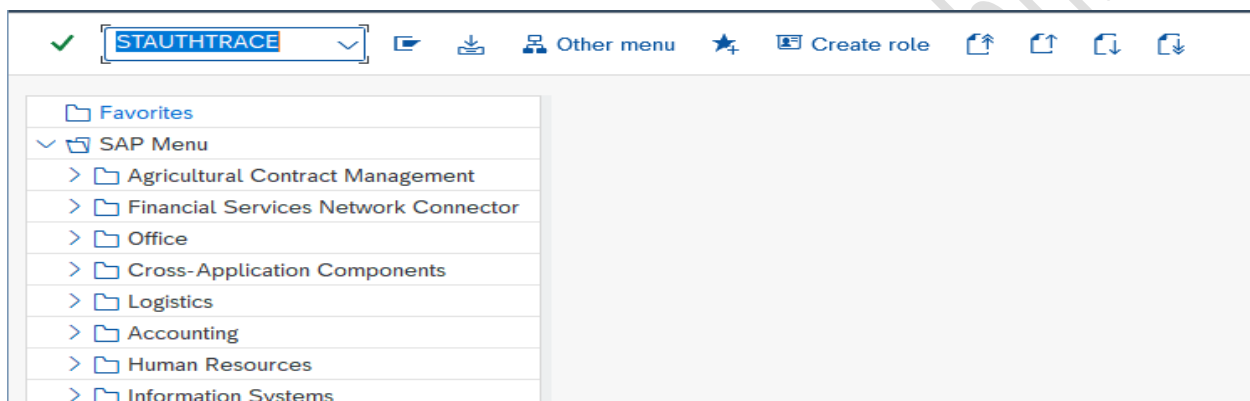
- ⇒ Exclusive **authorization** Trace.
- ⇒ Trace is applied across all **application servers**.

Note = > This T code is available in the updated version of SAP.

How to Use STAUTHTRACE:

✳ Access the Transaction:

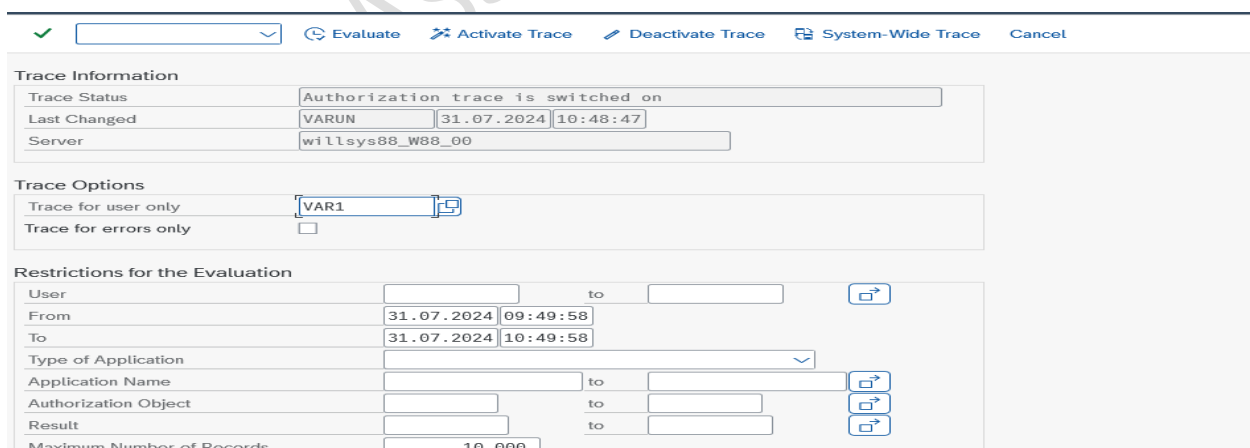
- Enter **STAUTHTRACE** in the SAP command field and press Enter.



The screenshot shows the SAP command field with 'STAUTHTRACE' entered. Below the command field is a menu bar with options like 'Favorites', 'SAP Menu', 'Agricultural Contract Management', 'Financial Services Network Connector', 'Office', 'Cross-Application Components', 'Logistics', 'Accounting', 'Human Resources', and 'Information Systems'.

✳ Start the Trace:

- Specify the **Username** and Click on "**Start Trace**" to begin recording **authorization checks**.



The screenshot shows the STAUTHTRACE configuration screen. It includes sections for 'Trace Information', 'Trace Options', and 'Restrictions for the Evaluation'. The 'Trace Information' section shows 'Authorization trace is switched on'. The 'Trace Options' section shows 'Trace for user only' set to 'VAR1'. The 'Restrictions for the Evaluation' section shows fields for 'User', 'From', 'To', 'Type of Application', 'Application Name', 'Authorization Object', 'Result', and 'Maximum Number of Records'.

✳ Perform Activities:

- Inform the **user** to perform the **actions** that are causing **access issues**, such as trying to **execute** a transaction they cannot access.

✳ Stop the Trace:

- Once the **activities** are completed, go back to **the STAUTHTRACE** screen, and click "**Deactivate Trace**".

Evaluate
Activate Trace
Deactivate Trace
System-Wide Trace
Cancel

✳ View and analyze results:

- Click on "**Display Trace**" to see the **results**. and review the **authorization**.

Search
Refresh
Print
Export
Import
User Buffer
Filter
Sort
Find
Copy
Paste
More

Date	Date/Time	User	Type	Application Name	Program Name	Che...Res...	Result	Addit...Ac	En...	Object	Field 1	Valu...	Field 2	Valu...	Field 3	Valu...	Fiel...	Valu...	Fie
31...	10:49:02...	VAR1	Tran...	PFCG	SAPLSUSE		4	Authorization che...		S_TCODE	TCD	PFCG							
31...	10:49:02...	VAR1	Tran...	SESSION_MANA...	/PERSONAS/CL_A...		12	No authorization i...		P_RUNTIME	P_APP_ID	P_AC...	01	P_AC...	01				
31...	10:49:02...	VAR1	Tran...	SESSION_MANA...	/PERSONAS/CL_A...		12	No authorization i...		P_RUNTIME	P_APP_ID	P_AC...	01	P_AC...					
31...	10:49:05...	VAR1	Tran...	SM37	SAPLSUSE		4	Authorization che...		S_TCODE	TCD	SM37							
31...	10:49:08...	VAR1	Tran...	SM36	SAPLSUSE		4	Authorization che...		S_TCODE	TCD	SM36							
31...	10:49:09...	VAR1	Tran...	SU01	SAPLSUSE		0	Authorization che...		S_TCODE	TCD	SU01							
31...	10:49:09...	VAR1	Tran...	SU01	SAPLSUSE		0	Authorization che...		S_USER_GRP	CLASS	ACTVT							
31...	10:49:09...	VAR1	Tran...	SU01	/PERSONAS/CL_A...		12	No authorization i...		P_RUNTIME	P_APP_ID	SU01	P_AC...	01	P_AC...	01			
31...	10:49:09...	VAR1	Tran...	SU01	/PERSONAS/CL_A...		12	No authorization i...		P_RUNTIME	P_APP_ID	P_AC...	01	P_AC...					

It displays the **details** about the **authorization objects**, **field values**, **program name** and the **return codes**.

Note = > Click on **System Wide Trace** to view the trace for each application server. Apply the necessary filters, then click on **Execute**.

Evaluate
Activate Trace
Deactivate Trace
Local Trace
Cancel

Trace Status

Logon Group

Server	Trace Status	User Name	Error...	Changed By	Date
☐ willsys88_W88_00	Undefined		☐		
☐ willsys88_W88_02	Undefined		☐		

Trace Options

Trace for user only

Trace for errors only ☐

Restrictions for the Evaluation

User to

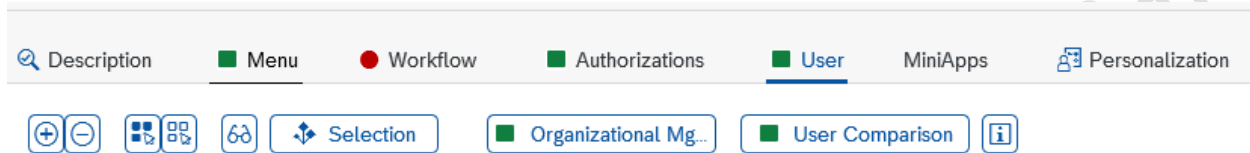
From

To

PFUD

Mass User Comparison

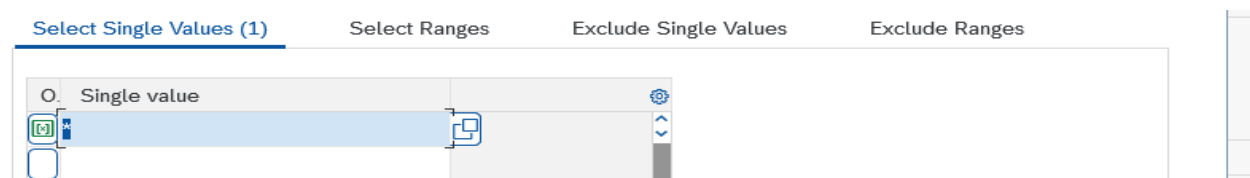
- ✓ For a **single role** we can do **user comparison** in PFCG -> User comparison Tab.



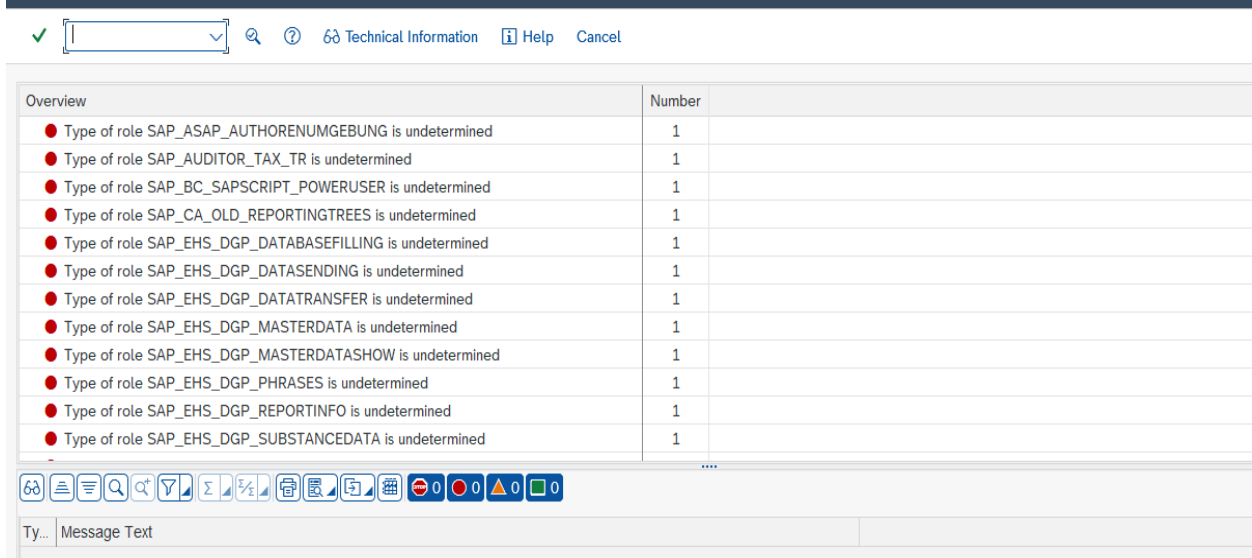
- ✓ If there is a requirement to perform **User comparison for 100 Roles** (Multiple roles), then we go to a T code called **PFUD**, to perform that.

A screenshot of the PFUD (Profile User Comparison) dialog box. The dialog has a title bar with a green checkmark and a dropdown menu. Below the title bar, there are several sections: 'Selection of roles' with a text input field and a button; 'Comparison Type' with three checked checkboxes: 'Profile matchup', 'Composite Indirect Assignments from Composite Roles', and 'HR Organizational Management: Reconciliation'; 'Options' with three checkboxes: 'Cleanups' (unchecked), 'Display Log' (checked), and 'Replicate Local HR Assignments in Central CUA' (unchecked); and 'Selection of action' with two radio buttons: 'Schedule or Check the Background Job for the Full Comparison' (unchecked) and 'Dialog Processing' (checked).

- ✓ Click from **multiple role selection** and copy from **clipboard**.



✓ Then, click on **execute**.



Overview	Number
● Type of role SAP_ASAP_AUTHORENUMGEBUNG is undetermined	1
● Type of role SAP_AUDITOR_TAX_TR is undetermined	1
● Type of role SAP_BC_SAPSCRIPT_POWERUSER is undetermined	1
● Type of role SAP_CA_OLD_REPORTINGTREES is undetermined	1
● Type of role SAP_EHS_DGP_DATABASEFILLING is undetermined	1
● Type of role SAP_EHS_DGP_DATASENDING is undetermined	1
● Type of role SAP_EHS_DGP_DATATRANSFER is undetermined	1
● Type of role SAP_EHS_DGP_MASTERDATA is undetermined	1
● Type of role SAP_EHS_DGP_MASTERDATASHOW is undetermined	1
● Type of role SAP_EHS_DGP_PHRASES is undetermined	1
● Type of role SAP_EHS_DGP_REPORTINFO is undetermined	1
● Type of role SAP_EHS_DGP_SUBSTANCEDATA is undetermined	1

What all important actions are performed in PFUD?

1) Profile matchup

- It ensures **profile** get assigned to user and the **role status is green** under **SU01**.

2) Composite role reconciliation

- Whenever **composite role is modified**, the role is **added or removed** from composite role, should reflect **to user as well**.

3) HR organizational management: reconciliation

- It ensures that **role** assigned to **position** or **org unit** of the user.

4) Cleanups

- It ensures that **expired roles** are removed from **SU01**, record of users.

Program related to **PFUD** = > **PFCG_TIME_DEPENDENCY**

SM01 = > Lock T code

EWZ5 = > Mass User Lock

SU10 = > SU12